Name: ________ Cohort/Batch: ________ Date: ________ Score: ________

MICROSOFT SQL SERVER PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A Databases

TOTAL: 10 QUESTIONS

Q1 What type of database is Microsoft SQL Server and what are its core strengths? Data Model

Q2 How does Microsoft SQL Server guarantee consistency and handle transactions? Architecture

Q3 What index types does Microsoft SQL Server support and when do you use each? Configuration

Q4 How do you design a schema or data model in Microsoft SQL Server? Hardening

Q5 How does Microsoft SQL Server scale horizontally? SSO / IdP

Q6 How do you monitor and tune slow queries in Microsoft SQL Server? Observability

Q7 What backup and restore strategies does Microsoft SQL Server support? Lifecycle

Q8 How do you handle schema migrations in Microsoft SQL Server? Compliance

Q9 What security features does Microsoft SQL Server provide? Testing

Q10 What are common anti-patterns when using Microsoft SQL Server in production? Tuning

ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 Microsoft SQL Server is a relational, document,

Mitigation

Microsoft SQL Server is a relational, document, key-value, graph, or time-series store. Its strengths such as ACID guarantees, horizontal scale, or flexible schema guide the workloads it is best suited for.

A2 Microsoft SQL Server provides either full ACID

Primitives

Microsoft SQL Server provides either full ACID transactions or eventual consistency depending on its CAP theorem positioning. Transaction support varies from none (simple K/V stores) to serializable isolation (relational DBs).

A3 Microsoft SQL Server offers B-tree, hash, full-text,

Hardening

Microsoft SQL Server offers B-tree, hash, full-text, spatial, or composite indexes depending on the engine. Choose based on query patterns: B-tree for range queries, hash for equality lookups, full-text for search.

A4 Schema design in Microsoft SQL Server involves

Gotchas

Schema design in Microsoft SQL Server involves normalizing relations (relational DB) or denormalizing for access patterns (document/key-value). Understanding query needs upfront prevents costly migrations later.

A5 Microsoft SQL Server scales via replication (read

Federation

Microsoft SQL Server scales via replication (read replicas) for read-heavy workloads and sharding (partitioning data by key) for write-heavy ones. Some Microsoft SQL Server implementations handle this automatically; others require manual configuration.

A6 Microsoft SQL Server provides a slow query

Observability

Microsoft SQL Server provides a slow query log, explain/explain plan, or profiling tools to surface inefficient queries. Common fixes include adding indexes, rewriting queries, or increasing cache size.

A7 Microsoft SQL Server supports logical backups (export

Automation

Microsoft SQL Server supports logical backups (export SQL or BSON), physical backups (filesystem snapshot), and continuous replication to a standby. Point-in-time recovery requires WAL/oplog archiving on top of backups.

A8 Schema migrations in Microsoft SQL Server are

Governance

Schema migrations in Microsoft SQL Server are managed with versioned migration files applied in order by a migration tool. Migrations should be idempotent and tested in staging before production to avoid downtime.

A9 Microsoft SQL Server supports role-based access control,

Assessment

Microsoft SQL Server supports role-based access control, encrypted connections (TLS), encryption at rest, and audit logging. Always restrict user privileges to the minimum needed and disable anonymous access in production.

A10 Common mistakes include selecting all columns instead

Optimization

Common mistakes include selecting all columns instead of needed fields, missing indexes on frequently queried columns, running migrations without backups, and storing large blobs directly in the database instead of object storage.